

物理宇宙学基础作业

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1.(a) 波长为:

$$\lambda_0 = \frac{\lambda}{1+z} = 150 \text{ nm}$$

观测者与类星体之间的共动距离为:

$$r = \int_0^z \frac{cdz'}{H_0 \sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}} = 6554.228 \text{ Mpc}$$

由于宇宙膨胀, 接收到的频率 ν 变为发射时的 $\frac{1}{1+z}$ 倍:

$$F_\nu d\nu = \frac{L_{(1+z)\nu}}{4\pi d_L^2} d(1+z)\nu, \quad d_L = r(1+z)$$

则类星体光度为:

$$L = 4\pi r^2 (1+z) F_{\text{obs}} = 2.056 \times 10^{23} \text{ W/Hz}$$

1.(b) 星系直径:

$$D = \frac{r\delta\theta}{1+z} = 39.7 \text{ kpc}$$

1.(c) 气体云与类星体间的距离为:

$$r_{\text{gas}} = \frac{r\delta\theta_{\text{gas}}}{1+z} = 79.4 \text{ kpc}$$

由于气体云与类星体之间为引力束缚系统, 不考虑宇宙学红移, 则观测到的波长为:

$$\lambda_{\text{gas}} = \lambda_0 = 150 \text{ nm}$$

亮度为:

$$F_{\text{gas}} = \frac{L}{4\pi r_{\text{gas}}^2} = 2.7 \times 10^{-21} \text{ W/(m}^2 \cdot \text{Hz)}$$

1.(d) 类星体发出的光传播到气体云所需的时间为:

$$\Delta t_{\text{gas}} = \frac{r_{\text{gas}}}{c} = 2.591 \times 10^5 \text{ yr}$$

在地球上观测, 时间膨胀:

$$\Delta t = \Delta t_{\text{gas}}(1+z) = 1.04 \times 10^6 \text{ yr}$$

即在约 104 万年后能观测到气体云的亮度变化。

1.(e) 设 Doppler 效应对红移的贡献为 z_v , 则:

$$\lambda = (1+z)(1+z_v)\lambda_0 = (1+z+\sigma_z)\lambda_0$$

于是 $\sigma_z = (1+z)z_v$, 即:

$$z_v = \frac{\sigma_z}{1+z} = 2.5 \times 10^{-4}$$

速度弥散为:

$$\sigma_v = cz_v = 75 \text{ km/s}$$

1.(f) 由题意 $\Delta r = 100 \text{ Mpc}$, 考虑:

$$r + \Delta r = \int_0^{z+\Delta z} \frac{cdz'}{H_0 \sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}} = 6654.228 \text{ Mpc}$$

数值计算可以得到: $\Delta z = 0.1$

而角距离:

$$\Delta\theta = \frac{(1+z)\Delta r}{r} = 0.061 \text{ rad} = 3.5^\circ$$